

Does occluding the ear free loudness budget for high-frequency gain? A modeling study of standard audiograms

Mark Shaver, Ph.D.

Department of Communication Sciences and Disorders

Wichita State University

Wichita, KS, USA

Corresponding author: Mark Shaver, Ph.D.

517 Oakwood St.

Rose Hill, KS 67133

Phone: 1-316-208-9588

Email: mark.shaver@wichita.edu

Date of submission: *[date]*

Prior presentation: None

Acknowledgments / funding: No grant support was received for this work; see Acknowledgments section.

Abstract

Background: In precipitous high-frequency hearing loss, restoring high-frequency audibility is difficult, usually attributed to the fitting rationale or to physiological damage.

Purpose: To test whether unamplified speech in the residual low frequencies consumes much of the loudness budget before gain is prescribed, and whether freeing that budget with negative low-frequency gain improves predicted intelligibility.

Research Design: Computational modeling with the AUDMOD specific-loudness model and the Speech Intelligibility Index (SII).

Study Sample: Ten standard audiograms (Bisgaard et al.) and sixteen synthetic audiograms.

Intervention: Relaxing the minimum insertion-gain floor from 0 to -10 dB, which requires an unvented earmold.

Data Collection and Analysis: The loudness budget was decomposed at 50, 65, and 80 dB SPL. At 65 dB SPL, gains were optimized at both floors in a sweep of audible edge, slope, and loudness budget and in an iso-loudness control matched to NAL-NL2, maximizing either the Johnson and Dillon desensitized SII or the uncorrected ANSI SII shown by real-ear systems.

Results: At 65 dB SPL, unamplified speech consumed 72% and 67% of the normative ceiling for the two standard audiograms with near-normal hearing through 2000 Hz, and 34% or less for the others. At NAL-NL2 loudness, relaxing the floor did not change the desensitized SII beyond search variability (about 0.008) for any standard audiogram, and at normal-hearing loudness no solution used negative gain. A benefit appeared only for synthetic audiograms with

near-normal hearing through 1000 Hz at the tightest budget (0.011 to 0.016 across seeds); there, fittings tuned to the ANSI SII gained 0.079 to 0.087 by that metric but 0.013 to 0.014 under the correction.

Conclusions: Near-normal low-frequency hearing consumes most of the loudness budget before gain is applied. For representative audiograms fitted at NAL-NL2 loudness, recovering that budget with negative low-frequency gain adds no predicted intelligibility, although the uncorrected SII displayed during verification can suggest otherwise.

Clinical Relevance Statement: For listeners with near-normal low-frequency hearing, open fittings cost no predicted intelligibility at typical prescribed loudness. A rise in the real-ear SII from occluding such a patient should not be read as real benefit, because the displayed index omits desensitization.

Key Words: hearing aids; hearing loss, high-frequency; loudness perception; speech intelligibility; computer simulation

Abbreviations: AI = artificial intelligence; AMT = Auditory Modeling Toolbox; ANSI = American National Standards Institute; HL = hearing level; LTASS = long-term average speech spectrum; NAL-NL2 = National Acoustic Laboratories nonlinear prescription, version 2; SII = Speech Intelligibility Index; SPL = sound pressure level

Introduction

In the fitting of precipitous or profound high-frequency hearing loss, achieving adequate high-frequency audibility without exceeding normative loudness targets is a central clinical challenge. Historically, the failure to restore high-frequency audibility has been attributed to the inherent physiological damage of the auditory periphery—specifically, hearing-loss desensitization and dead regions (Moore, 2001), which sharply limit the benefit of high-frequency audibility where thresholds exceed about 55 dB HL (Ching et al., 1998; Hogan and Turner, 1998). Consequently, modern prescriptive rationales like NAL-NL2 (Keidser et al., 2011) deliberately limit high-frequency gain for profound thresholds to avoid prescribing “wasted” amplification.

Desensitization limits the usefulness of high-frequency gain; loudness may limit how much of it can be provided. Unamplified speech produces a baseline loudness (L_0) through residual hearing. If a patient’s target loudness is held to a normative ceiling (L_{cap}), the loudness left to “purchase” high-frequency audibility through gain is $L_{cap} - L_0$. Because this budget is set by the audiogram and the speech spectrum rather than by the fitting rationale, it could limit high-frequency gain under any loudness-based rationale (e.g., NAL-NL2) and under any attempt to maximize intelligibility.

The ANSI S3.5-1997 Speech Intelligibility Index (SII) (ANSI, 1997) assigns most of its importance weight to the 1–4 kHz bands and little below 500 Hz, so an algorithm that maximizes it under a strict loudness budget will tend to attenuate low-frequency audibility to fund high-frequency gain. Relaxing a constraint cannot lower the optimum of an optimization problem, so the clinical relevance lies in quantifying the magnitude of this trade-off for realistic audiometric

profiles, and in whether it survives when the index is corrected for desensitization. Ching et al. (2001) demonstrated that fitting options should be compared at equal loudness and with an index corrected for desensitization, the principle on which NAL-NL1 and NAL-NL2 were derived. The present study applies that principle to a question those comparisons did not address: whether the loudness of unamplified speech in near-normal low frequencies, which a 0 dB minimum gain cannot reduce, limits the budget left for high-frequency gain, and whether recovering it with negative low-frequency gain, which requires an unvented fitting, improves predicted intelligibility. It also asks whether any such limit is peculiar to precipitous losses or is a general property of near-normal low-frequency hearing.

Methods

Study Design

The study comprised two analyses. The first, a loudness budget decomposition, required no gain optimization: it compared the loudness of unamplified speech with a normative ceiling at 50, 65, and 80 dB SPL to show how much of the budget residual hearing consumes and how that proportion changes with input level. The second, which tested whether relaxing the minimum insertion-gain floor improves intelligibility within that budget (the floor effect: the SII with a -10 dB floor minus the SII with a 0 dB floor), required optimizing gains and was performed at 65 dB SPL only, the level of average conversational speech. The audiogram family sweep and the iso-loudness control therefore describe behavior at 65 dB SPL; behavior at other levels is considered in the Limitations.

The AUDMOD Specific-Loudness Model

Loudness was computed with the AUDMOD specific-loudness model (Nielsen and Bramsløw, 1993; Bramsløw, 2004). AUDMOD was selected because it is the only loudness model in the Auditory Modeling Toolbox (Majdak et al., 2022) built specifically to account for hearing loss while retaining the excitation-to-specific-loudness compressive transformation. The model of Chen et al. (2011) also addresses cochlear hearing loss but is not implemented for hearing loss in that toolbox, and a C++ port validated stage by stage against a reference implementation (below) was needed for the number of loudness evaluations the optimization requires. This model converts acoustic excitation patterns into specific loudness (sones/ERB) and integrates them across the frequency spectrum to predict overall broadband loudness. Passing the unamplified long-term average speech spectrum (LTASS) through it gives the baseline loudness L_0 produced by residual hearing. Throughout this work the LTASS is the ANSI S3.5-1997 standard speech spectrum for normal vocal effort (ANSI, 1997), scaled to the evaluation level; the SII uses it in its critical bands, and the loudness model receives it sampled at the six octave frequencies and interpolated (Supplementary Section S2).

Study Sample and Audiometric Profiles

The study used the ten standard audiograms of Bisgaard et al. (2010), proposed for the IEC 60118-15 hearing aid measurement procedure and derived from a clinical database of measured audiograms: seven flat and moderately sloping configurations (N1–N7, very mild to profound) and three steep sloping configurations (S1–S3), all sensorineural (Table 1). Thresholds at the octave frequencies from 250 to 4000 Hz were taken from their published tables; because those tables end at 6000 Hz, the 8000 Hz threshold was set equal to the 6000 Hz value. Thresholds are

in dB HL for supra-aural earphones, the reference assumed by the AUDMOD loudness model; the SII uses the same values through the reference internal noise of ANSI S3.5-1997.

Table 1. Standard Audiograms (dB HL)

Profile	Category	250	500	1000	2000	4000	8000
N1	Very mild	10	10	10	15	30	40
N2	Mild	20	20	25	35	45	50
N3	Moderate	35	35	40	50	60	65
N4	Moderate/severe	55	55	55	65	75	80
N5	Severe	65	70	75	80	80	80
N6	Severe	75	80	85	90	100	100
N7	Profound	90	95	105	105	105	105
S1	Very mild, steep	10	10	10	15	55	70
S2	Mild, steep	20	20	25	55	95	95
S3	Mod./severe, steep	30	35	60	75	80	85

Note: From Bisgaard et al. (2010), Tables 2 and 4. The 8000 Hz value repeats their 6000 Hz value.

The Open-NL Computational Instrument

Gains were optimized with Open-NL, an open-source computational testbed that couples a multi-start Nelder-Mead SII optimizer to the specific-loudness engine. Open-NL serves here as an analytical instrument, not a clinical prescription. Its wide dynamic range compression was bypassed, so the optimizer solved for linear insertion gain at 65 dB SPL.

The optimizer was subject to a minimum insertion-gain floor in every band. A negative floor represents the passive insertion loss of an occluding earmold (the real-ear occluded response minus the real-ear unaided response). Measured insertion loss is frequency dependent and smallest at the lowest frequencies, even for unvented molds (Kuk et al., 2009), so the flat

–10 dB floor used here is an idealization that is likely optimistic in the lowest bands. For these audiograms the floor binds, if at all, only at the low frequencies.

Comparison targets were generated with the NAL-NL2 software (v2.0, dll v2.15, Clinician Edition) (Keidser et al., 2011) from the same six thresholds in each ear, for an adult male experienced bilateral user of a non-tonal language, with dual (adaptive) compression speed, 18 channels, the default compression threshold, supra-aural earphones as the threshold transducer, and an occluded earmold with #13 tubing. Male was chosen to match Open-NL’s heuristic prescription, and the occluded coupling keeps the targets free of venting effects. Real-ear insertion-gain targets for speech at 65 dB SPL were used. The bilateral setting applies a binaural loudness correction that lowers prescribed gain, whereas the loudness model used here is monaural. This mismatch lowers NAL-NL2’s predicted loudness, and therefore the loudness cap of the iso-loudness control, by an amount that was not quantified.

Loudness Model Implementation and Verification

Open-NL evaluates tens of thousands of candidate gain curves per optimization, and calling the AUDMOD implementation in AMT 1.6.0 (bramslow2004) (Majdak et al., 2022) through Octave for each evaluation was prohibitively slow. The model was therefore ported to C++ using Rcpp. Across 23 test cases, the port matched the AMT implementation at every processing stage to within a maximum relative discrepancy of 5.71×10^{-15} ; this establishes that the port reproduces AUDMOD, not that AUDMOD itself is accurate. Reference-free checks confirmed that loudness is zero below threshold and that recruitment is present and ordered by threshold, and in normal hearing loudness grew by a factor of 1.75 to 1.91 per 10 dB above 40 phons, mildly compressive

relative to the textbook doubling. The absolute sone scale is inherited from AUDMOD and has not been independently verified against loudness judgments. Checks of the input path and further details are given in Supplementary Section S1.

Intelligibility Metrics

Two versions of the SII were used. The *ANSI SII* is the index as standardized (ANSI, 1997), with no correction for desensitization. The *complete* desensitized SII (hereafter the complete SII) reduces each band’s audibility K_i by a desensitization factor that falls as the band’s sensorineural threshold T_i (dB HL) rises, $m_i = 1/(1 + e^{0.075(T_i - 66)})$, using the published Johnson and Dillon (2011) correction, a power mean:

$$K'_i = (K_i^{p_i} + m_i^{p_i})^{1/p_i}, \quad p_i = T_i/8 - 15.$$

Because p_i is strongly negative at mild to moderate thresholds ($p_i = -10$ at 40 dB HL), this behaves approximately as $\min(K_i, m_i)$ there: once a band’s audibility reaches m_i , further audibility earns little credit. At severe to profound thresholds p_i approaches -1 and the correction is more gradual. The complete SII is the optimizer’s objective. The ANSI SII, which real-ear measurement systems typically display, is reported for the same solutions.

Optimization Procedure

Gains were optimized by multi-start Nelder-Mead simplex search over six free parameters: a gain shift at each octave frequency from 250 to 8000 Hz, added to Open-NL’s built-in rule-based heuristic prescription and interpolated logarithmically across the 21 analysis bands. The heuristic used the package defaults: a fully occluded coupling with no vent correction, the severe-loss booster disabled, and no demographic adjustments.

The solutions are penalized-optimal rather than strictly SII-optimal: the optimizer maximizes the complete SII, multiplied by 100, minus a set of penalty terms, so that one objective point corresponds to 0.01 SII. Three penalties shape the solutions reported here. The anchor penalty subtracts 0.1 point for every decibel by which the gain at an octave frequency is shifted, in either direction, from the heuristic prescription, pulling solutions toward it. The roughness penalty subtracts 0.001 point per squared decibel of difference between the gains at adjacent octave frequencies, discouraging jagged responses. The loudness penalty subtracts 2000 points per sone by which aided loudness exceeds the cap, which in practice makes the cap a hard ceiling. The remaining penalties are safeguards that were inactive or rarely engaged here (Supplementary Section S2).

Each optimization used 20 random restarts unless stated otherwise, and the solver retains the solution with the best penalized objective, which need not have the highest SII. The random seed is fixed by the audiogram and level, so a 40-restart run repeats the first 20 restarts and can only improve on the 20-restart objective. Because repeated runs are therefore identical, search variability was estimated separately by repeating key comparisons with five further seeds (Supplementary Section S4).

The anchor penalty keeps solutions close to a plausible prescription shape, but it also works against the manipulation under test. Because the heuristic is floored at 0 dB before shifts are applied, any gain below 0 dB is a penalized shift: driving two low-frequency bands to the -10 dB floor costs $2 \times 10 \text{ dB} \times 0.1 = 2$ points, or 0.02 SII, the same order as the floor effects of interest. With the anchor penalty on, the floor effect is therefore biased downward. The tightest-budget

sweep and the iso-loudness control were repeated with the anchor weight set to zero (20 restarts). These anchor-off solutions are constrained only by the loudness cap, the floor, and the weak roughness penalty; they indicate what an essentially unrestricted SII maximizer could achieve rather than realistic prescriptions. Anchor-on and anchor-off results are reported together as a bracket.

Results

Loudness Budget Decomposition

The unamplified loudness of speech (L_0) was computed at 50, 65, and 80 dB SPL and compared with a normative ceiling (L_{cap}), defined as the loudness of unaided speech for a 0 dB HL listener at the same level: 3.16, 9.03, and 22.39 sones, respectively. The ceiling is the same for every listener. The difference, $L_{cap} - L_0$, is the model-predicted budget available to “purchase” high-frequency audibility through gain before aided loudness exceeds that of normal hearing.

An empirical anchor for this ceiling would be preferable, and NAL-NL2 was examined as a candidate. It does not provide one: NAL-NL2-prescribed aided loudness for 65 dB SPL speech, as a fraction of the ceiling, ranges from 0.07 to 0.78 across nine of the standard audiograms (0.78 for N1, 0.74 for S1, 0.54 for N2, 0.49 for S2, and 0.26 or less for the rest) and is zero for N7, because NAL-NL2 optimizes loudness against its own criteria rather than a fixed proportion of normal-hearing loudness.

L_{cap} is therefore a modeling assumption, not a measured quantity, and target loudness varies across patients and rationales. The floor analyses below accordingly do not rely on it alone:

the audiogram family sweep sets each budget relative to the listener’s own unaided loudness, at four values, and the iso-loudness control uses the loudness NAL-NL2 prescribes.

Table 2 presents this decomposition for the ten standard audiograms.

Table 2. Loudness Budget Decomposition: Unamplified Loudness (L_0 , Sones) and Fraction of the Normative Ceiling Consumed, at 50, 65, and 80 dB SPL

Profile	L_0 (sones)			Ceiling consumed (%)		
	50	65	80	50	65	80
N1	1.76	6.50	17.73	55.9	72.0	79.2
N2	0.44	3.09	10.43	13.8	34.2	46.6
N3	0.00	0.60	4.11	0.0	6.6	18.4
N4	0.00	0.00	1.16	0.0	0.0	5.2
N5	0.00	0.00	0.00	0.0	0.0	0.0
N6	0.00	0.00	0.00	0.0	0.0	0.0
N7	0.00	0.00	0.00	0.0	0.0	0.0
S1	1.71	6.06	16.29	54.3	67.1	72.8
S2	0.43	2.71	8.63	13.6	30.0	38.6
S3	0.00	0.42	2.61	0.0	4.7	11.7

Note: The ceiling L_{cap} is 3.16, 9.03, and 22.39 sones at 50, 65, and 80 dB SPL.

The fraction consumed depends mainly on how close the low-frequency thresholds are to normal. The two audiograms with thresholds of 10 to 15 dB HL through 2000 Hz, N1 and S1, consume 72.0% and 67.1% of the ceiling at 65 dB SPL, although S1 has a steep high-frequency loss and N1 only a mild one. With low-frequency thresholds of 20 to 25 dB HL, as in N2 and S2, the fraction falls to 34.2% and 30.0%, even though S2 has the greatest overall high-frequency slope in the set. From N3 and S3 upward, unamplified speech consumes 6.6% or less, and for N4 to N7

it is inaudible to the model at 65 dB SPL. The fraction consumed rises with input level wherever speech is audible (Figure 1), so the constraint tightens as speech gets louder: N1 consumes 55.9%, 72.0%, and 79.2% of the ceiling at 50, 65, and 80 dB SPL. If a prescription enforces a minimum insertion gain of 0 dB, these L_0 values cannot be reduced, and only the remainder of the ceiling is available for high-frequency gain. Whether relaxing the floor to recover part of L_0 improves intelligibility was tested at 65 dB SPL in the analyses that follow.

The Audiogram Family Sweep

Sixteen synthetic audiograms crossed four audible edge frequencies (1000, 1500, 2000, 3000 Hz) with four high-frequency slopes (20, 30, 40, 50 dB/octave): thresholds were 10 dB HL up to the audible edge and rose above it at the stated slope, to a maximum of 110 dB HL. Each was evaluated at four loudness budgets and two minimum-gain floors (0 and -10 dB), giving 64 paired cells. Unlike the fixed ceiling of the budget decomposition, each budget was set relative to the listener's own unaided loudness, at 0.5, 1, 2, or 3 sones above L_0 . For scale, NAL-NL2 at 65 dB SPL spends 0.54 to 2.04 sones above L_0 for the standard audiograms (Tables 2 and 4; N7 excepted). Each floor was optimized with 20 random restarts, and each solution was scored with the complete and ANSI SII.

At the tightest budget the loudness cap should bind at both floors. Fifteen of the 16 pairs matched in achieved loudness to within 0.001 sones; the sixteenth matched after re-optimization with 40 restarts, and its 40-restart solutions are used throughout. Overall, 50 of the 64 paired cells match (16, 11, 10, and 13 of 16 at budgets of 0.5, 1, 2, and 3 sones).

Because the 0 dB feasible set lies within the -10 dB set, relaxing the floor cannot lower the penalized objective at a true optimum, but it can lower the SII when the wider search space lets the solution trade a little SII for a smaller penalty. Across the 50 matched pairs, 23 complete-SII floor effects are negative (median magnitude 0.0009, maximum 0.0084). In all but five of the 50 the penalized objective was the same or higher at -10 dB, and in those five it was lower by at most 0.04 points (0.0004 SII), so the search itself converged. Complete-SII floor effects no larger in magnitude than the largest negative value (0.0084, hereafter about 0.008) are therefore treated as indistinguishable from zero; this is the resolution limit. An independent estimate agreed: when the floor comparison was repeated with six random seeds for the four 1000 Hz edge audiograms and for N1, N2, S1, and S2 at NAL-NL2 loudness, the pooled within-case standard deviation of the floor effect was 0.0037 (2 SD = 0.0075; Supplementary Section S4).

Table 3. Audiogram Family Sweep: Mean Floor Effect (SII at -10 dB minus SII at 0 dB) at the 0.5-Sone Budget, by Audible Edge, with the Anchor Penalty On and Off

Audible edge	Complete (on)	Complete (off)	ANSI (on)	ANSI (off)
1000 Hz	+0.014	+0.015	+0.039	+0.047
1500 Hz	0.000	+0.004	+0.004	+0.017
2000 Hz	+0.002	+0.002	+0.005	+0.010
3000 Hz	-0.003	+0.001	-0.003	+0.004

Note: On and off indicate whether the optimizer's anchor penalty was applied (see Optimization Procedure). Each value is the mean over the four slopes. All 16 anchor-on pairs matched in loudness; 8 of the 16 anchor-off pairs did.

Table 3 gives the floor effect at the tightest budget, averaged over the four slopes at each audible edge. With the anchor penalty on, the effect at the 1000 Hz edge is 0.014 under the

complete SII and 0.039 under the ANSI SII. At the 1500, 2000, and 3000 Hz edges it is within the resolution limit under the complete SII and at most 0.005 under the ANSI SII. The benefit at the 1000 Hz edge does not depend on the random starts: across six seeds, all 24 floor effects for those four audiograms were positive (0.005 to 0.020; audiogram means 0.011 to 0.016). The effect disappears as the budget loosens: at 1, 2, and 3 sones no edge shows a mean benefit above 0.003 under either metric.

Removing the anchor penalty leaves the complete-SII floor effect nearly unchanged (0.015 at the 1000 Hz edge and at most 0.004 elsewhere) but raises it under the ANSI SII (0.047 at the 1000 Hz edge and 0.017 at 1500 Hz). Only 8 of the 16 anchor-off pairs match in loudness. In four (edges of 1500 to 3000 Hz, slopes of 20 to 40 dB/octave), the -10 dB solution leaves 0.83 to 1.38 sones of the budget unspent while changing the complete SII by at most 0.005.

To locate the source of the benefit at the 1000 Hz edge, each -10 dB solution (anchor on) was rescored with only its low-frequency (250–1000 Hz) or only its high-frequency (2000–8000 Hz) gains applied to the 0 dB solution. The low-frequency cut alone changed the complete SII by at most 0.0005, because those near-normal bands remained fully audible, whereas the high-frequency gains alone reproduced the entire improvement, in the bands centered at 1600 to 3400 Hz, where audibility is limited by threshold rather than by masking. Applied without the low-frequency cut, however, those high-frequency gains would have exceeded the loudness cap by 0.44 to 0.72 sones, so the 0 dB solution could not have reached them. The floor effect is thus a loudness-budget effect: low-frequency audibility the SII does not need is traded for audibility at higher frequencies that it does.

In the anchor-on solutions, the reallocation of gain at matched loudness can be read directly from the paired solutions. At the tightest budget and the 1000 Hz edge, averaged over the four audiograms, relaxing the floor lowers gain by 4.0 dB at 500 Hz and raises it by 7.2 dB at 2000 Hz; gain at 8000 Hz also falls, by 5.2 dB, where the complete correction credits little audibility at these thresholds. Because these pairs are at matched loudness, capacity drawn from the near-normal low frequencies and from 8000 Hz is spent around 2000 Hz (Figure 2).

Iso-Loudness Control

To isolate the SII cost of preventing low-frequency attenuation at a clinically realistic loudness, Open-NL was constrained to the loudness of the NAL-NL2 prescription for each profile at 65 dB SPL and optimized twice, with 0 dB and -10 dB floors. As in the sweep, it optimized the complete SII, and each solution, together with the NAL-NL2 prescription, was scored with both metrics. The control was repeated with the optimizer maximizing the ANSI SII, to represent a fitting tuned to the index that real-ear systems display. The aim is not to compare Open-NL with the clinical rationale but to isolate the cost of the 0 dB floor at matched loudness. Table 4 separates the *optimizer effect* (Open-NL at 0 dB minus NAL-NL2) from the *floor effect* (Open-NL at -10 dB minus 0 dB) under the complete SII; Table 5 gives the floor effect under both metrics, with the anchor penalty on and off.

Table 4. Iso-Loudness Control: Complete Desensitized SII (Constrained to NAL-NL2 Loudness at 65 dB SPL; Anchor Penalty On)

Profile	NAL-NL2 loudness (sones)	NAL-NL2 SII	Open-NL (0 dB)	Open-NL (−10 dB)	Optimizer effect	Floor effect
N1	7.04	0.883	0.900	0.899	+0.017	−0.001
N2	4.84	0.809	0.825	0.823	+0.016	−0.002
N3	2.32	0.643	0.648	0.645	+0.005	−0.003
N4	2.04	0.428	0.424	0.424	−0.005	0.000
N5	1.57	0.243	0.233	0.233	−0.010	0.000
N6	0.63	0.119	0.114	0.114	−0.005	0.000
N7	0.00	0.023	0.016	0.016	−0.007	0.000
S1	6.64	0.792	0.814	0.810	+0.022	−0.004
S2	4.42	0.558	0.594	0.595	+0.036	+0.001
S3	1.42	0.418	0.421	0.424	+0.003	+0.003

Table 5. Iso-Loudness Control: Floor Effect under the Complete and ANSI SII, with the Anchor Penalty On and Off

Profile	Complete (on)	Complete (off)	ANSI (on)	ANSI (off)
N1	-0.001	+0.004	0.000	+0.009
N2	-0.002	+0.001	-0.004	+0.005
N3	-0.003	-0.005	-0.003	0.000
N4	0.000	0.000	0.000	0.000
N5	0.000	0.000	0.000	0.000
N6	0.000	0.000	0.000	0.000
N7	0.000	0.000	0.000	0.000
S1	-0.004	0.000	-0.002	+0.012
S2	+0.001	0.000	0.000	0.000
S3	+0.003	+0.001	+0.002	+0.002

Note: On and off indicate whether the optimizer’s anchor penalty was applied (see Optimization Procedure). Solutions were optimized on the complete SII. Only N1 and S1 received negative gain, and N2 also without the anchor penalty; elsewhere the floor did not engage, and nonzero values reflect search variability.

With the anchor penalty on (Table 4), relaxing the floor changed the complete SII by -0.004 to +0.003 for every audiogram, within the resolution limit. Only N1 and S1, the two audiograms with near-normal hearing through 2000 Hz, received any negative gain (at 500 and 1000 Hz), and for both the effect was slightly negative (-0.001 and -0.004); for N4 to N7 the floor never engaged. Without the anchor penalty the range was -0.005 to +0.004 (Table 5), and the ANSI SII of the same solutions changed by at most 0.012 (S1). At the loudness NAL-NL2 prescribes, none of the standard audiograms gained predicted intelligibility from relaxing the floor. For N4 to N7 unaided speech is inaudible to the model, so the floor could not matter; the

informative cases are the four with audible low-frequency speech, N1, N2, S1, and S2. Across six random seeds their mean floor effects were -0.003 to $+0.002$. Single runs for N1 and N2 varied more (-0.010 to $+0.010$; SD 0.006 to 0.007), so an individual value for these two is uncertain to about ± 0.01 , but no mean approached the benefit at the 1000 Hz edge.

The reasons differ across the set. N1 and S1 have the tightest budgets, NAL-NL2 spending only 0.54 and 0.58 sonas above their unaided loudness, but their thresholds stay near normal through 2000 Hz, like the sweep audiograms with edges of 2000 Hz and above, for which the floor effect was also within the resolution limit (Table 3). N2 and S2 consume less of the ceiling, and NAL-NL2 leaves them about 1.7 sonas of budget; from N3 and S3 upward there is almost no unaided loudness to recover. The optimizer effect was positive for near-normal or mild low-frequency loss (up to $+0.036$, S2) and slightly negative for N4 to N7 (-0.005 to -0.010), where the anchored search fell short of NAL-NL2; this does not affect the floor effect.

Fittings tuned to the ANSI SII behaved the same way. By that metric relaxing the floor gained at most 0.011 with the anchor penalty and 0.027 without it (S2), but under the complete correction the same solutions changed by -0.005 to $+0.008$ (Supplementary Section S2).

For the fittings optimized on the complete SII (Tables 4 and 5), all pairs matched in loudness to within 0.008 sonas with the anchor penalty. Without it, all but N2 matched to within 0.012 sonas; N2's -10 dB solution left 0.66 sonas of the budget unspent.

Floor Effect at the Normative Ceiling

The floor analyses above constrain loudness to NAL-NL2's prescription or to small increments above unaided loudness, never to L_{cap} itself. To test the comparison at the normative

ceiling, the optimization for N1, N2, S1, and S2 was repeated with the cap set to L_{cap} (9.03 sones), a budget of 2.5 to 6.3 sones above L_0 rather than the 0.5 to 1.8 sones NAL-NL2 uses. With or without the anchor penalty, no solution gave any band negative gain, so the floor never engaged; differences between floors reflected only search variability (largest 0.012, N2). Only N1 with the anchor penalty spent the whole budget; the other solutions stopped 0.1 to 3.8 sones short of it. The extra loudness raised the complete SII above that of NAL-NL2 by 0.007 to 0.031 (anchor on). When loudness is this loosely constrained, the optimizer has no reason to cut low-frequency gain (Supplementary Section S4).

Discussion

How much of the normal-hearing loudness ceiling remains for high-frequency gain is set largely by the low frequencies. For N1 and S1, the standard audiograms with near-normal hearing through 2000 Hz, unamplified speech at 65 dB SPL generates 6.50 and 6.06 sones, 72% and 67% of the 9.03-sone normal-hearing ceiling, before any gain is applied, although the exact proportion depends on the chosen L_{cap} . With a minimum insertion gain of 0 dB, the remaining quarter to third of the budget must cover the entire high-frequency prescription. The steepness of the high-frequency loss matters much less: S2, with the greatest overall high-frequency slope in the set, consumes 30% because its low-frequency thresholds are 20 to 25 dB HL.

This has a practical consequence. Any benefit from relaxing the floor relies on negative low-frequency insertion gain, but digital gain reduction cannot lower the ear-canal level below the direct sound path in an open fitting: an open earset provides no measurable insertion loss (Kuk et al., 2009), and venting reduces amplified low-frequency output rather than the unamplified

direct sound (Stuart et al., 1999). True negative low-frequency insertion gain therefore requires an unvented earmold (Kuk et al., 2009). Even unvented molds attenuate only modestly at the lowest frequencies, and they produce the largest occlusion effects of any configuration (Kuk et al., 2009; Kiessling et al., 2005), a particular burden for listeners with near-normal low-frequency hearing.

Conversely, an open or vented fitting, the standard of care when low-frequency hearing is near normal, limits the low-frequency insertion gain to about 0 dB or more. In the model this cost no predicted intelligibility for the standard audiograms fitted at NAL-NL2 loudness, although larger vents also reduce the gain available before feedback (Kuk et al., 2009), independently limiting high-frequency gain.

This analysis does not suggest that clinicians pursue these SII maxima by occluding patients and attenuating low frequencies. Because of desensitization (Ching et al., 1998; Hogan and Turner, 1998), the ANSI SII overpredicts speech recognition in severe high-frequency loss, although high-frequency amplification is rarely harmful: listeners with high-frequency cochlear dead regions still benefited from it on average, if less in noise (Cox et al., 2011; Pepler et al., 2016).

Which version of the SII is consulted also matters clinically. Real-ear verification systems report the ANSI SII without any correction for hearing-loss desensitization (Audioscan, 2026), and aided SII values are used to judge whether a fitting is acceptable (Audioscan, 2026; Scollie, 2018; Wiseman and McCreery, 2023). Where relaxing the floor helped at all, the uncorrected index overstated the benefit several-fold. At the 1000 Hz audible edge and the tightest budget, fittings tuned to the ANSI SII gained 0.079 to 0.087 from relaxing the floor by that metric but

0.013 to 0.014 under the complete correction, no more than fittings tuned to the corrected index itself (0.014 to 0.015); at the 1500 Hz edge a displayed gain of 0.027 to 0.038 corresponded to 0.004 to 0.005 (Supplementary Section S2). For the standard audiograms, fittings tuned to the ANSI SII showed up to 0.027 by that metric and no change beyond the resolution limit under the correction. Ching et al. (2001) cautioned that simple audibility indices and visual displays of audibility, which ignore desensitization, can lead to erroneous judgments about the relative benefits of hearing aid options, and fitting to maximize the uncorrected articulation index lowered scores for listeners with sloping high-frequency loss and near-normal low-frequency hearing (Rankovic, 1991). A rise in the displayed SII from occluding such a patient should therefore not be taken as evidence of benefit, and the results support continued venting to manage occlusion rather than pursuit of ANSI SII improvements.

Taken together, these analyses separate two explanations for the difficulty of restoring high-frequency audibility when low-frequency hearing is near normal. Unamplified speech in near-normal low frequencies does consume most of the normal-hearing loudness ceiling before any gain is prescribed. But recovering part of it by relaxing the minimum-gain floor bought no predicted intelligibility under the desensitized index for representative audiograms at the loudness NAL-NL2 prescribes, with or without the anchor penalty, and at the full normal-hearing ceiling no solution cut low-frequency gain at all. A benefit required a combination absent from the standard set: near-normal hearing through 1000 Hz, a steep loss immediately above it, and a tight budget. Even there it was small under the published correction (0.011 to 0.016), and it came from gain above the low frequencies that the loudness cap could not otherwise accommodate,

funded by a low-frequency cut. Because negative low-frequency insertion gain requires an unvented fitting, these results give no reason to sacrifice venting. Strategies that deliver high-frequency cues to regions of better hearing, such as frequency lowering (Simpson, 2009), remain the more plausible route; the present analysis does not model them, and their benefit under a loudness budget remains to be tested.

Limitations

Several constraints apply to these computational results. The loudness model is monaural and does not capture binaural loudness summation in impaired listeners. It applies stationary loudness integration to a static long-term average speech spectrum, whereas real speech fluctuates and compression acting on it may yield different instantaneous loudness (Souza, 2002). The analysis is an index-level optimization without behavioral validation. Finally, it used idealized thresholds (the ten standard audiograms of Bisgaard et al. (2010), which summarize a clinical database, and sixteen synthetic sweep audiograms) and a single speech spectrum; individual patients would require measured audiograms and varied speech inputs. The standard set contains no audiogram with near-normal hearing through 1000 Hz and a steep loss immediately above, the configuration for which the sweep found a benefit. Steeply sloping losses are uncommon (Bisgaard et al., 2010), but for them the conclusion rests on the synthetic sweep.

The SII was computed in quiet. In noise, low-frequency attenuation also changes how much noise reaches the ear and how much it masks upward, and an open fitting passes environmental noise directly; these effects, like the occlusion effect, own-voice quality, loss of the open-ear resonance, and feedback, are outside the model.

The floor analyses were also conducted only at 65 dB SPL with linear gain. Figure 1 shows the budget tightening as input level rises, which would favor a larger floor effect for loud speech; however, additional 2–4 kHz gain already lowered the SII slightly at 65 dB SPL, and the SII’s level-distortion factor penalizes high presentation levels more strongly. The direction of the effect at 50 and 80 dB SPL therefore cannot be inferred from these results and would require a level-specific (compression) analysis.

The flat –10 dB floor likely overstates achievable attenuation in the lowest bands, so the floor effects are likely optimistic estimates of the benefit of passive attenuation. A frequency-shaped floor based on measured insertion loss would be more realistic, although measured attenuation depends strongly on earmold seal, insertion depth, and slit leak.

The anchor penalty biases the floor effects in the opposite direction, and its size was tested directly. The anchor-off solutions indicate what an essentially unpenalized SII maximizer could achieve but are weakly regularized and are not realistic prescriptions. The anchor-off solutions remove the anchor’s downward bias but keep the optimistic flat floor, so apart from incomplete convergence they are an upper estimate of the floor effect; the anchor-on solutions carry both biases, which act in opposite directions. The anchor-off solutions are also not exact optima: in the four sweep pairs that left budget unspent, additional 8 kHz gain would still have raised the SII by 0.002 to 0.003.

The two models also disagree near threshold. For the profound audiogram N7, AUDMOD predicts zero loudness even for the NAL-NL2 fitting, so the iso-loudness cap is zero and that comparison is degenerate, whereas the SII still credits a little audibility (0.07 for the ANSI SII). The

models reference threshold differently, AUDMOD through supra-aural earphone reference levels and the SII through the reference internal noise of ANSI S3.5-1997, so speech near threshold can count as audible to one and not the other. The floor analyses concern listeners with audible low-frequency speech, for whom the discrepancy does not arise.

Acknowledgments

Generative AI tools were used in preparing this work. Gemini 3.1 Pro (Google DeepMind) served as a programming and copyediting assistant, refactoring R and C++ code and editing the manuscript text. Claude (Anthropic; Opus 5.5 and Sonnet 5 models) was used to audit the analysis code and its outputs, to write diagnostic and analysis scripts in R, to draft portions of the text, including the abstract, and to locate and verify references against publisher records. All reported analyses were run by the author. The author verified all AI-assisted code through numerical checks against independent outputs, reviewed and edited all AI-assisted text, and takes full responsibility for the content, including the accuracy of all references.

Conflict of Interest

The author declares no conflicts of interest.

Funding

This work received no external funding.

Data Availability

The code used to execute the computational simulations, reproduce the dataset, and generate the figures for this study is fully open-source and available on GitHub (<https://github.com/r-gregmisc/SII>).

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Figure Legends

Figure 1. Fraction of the normative loudness ceiling consumed by unamplified speech, for the ten standard audiograms (Table 1) at input levels of 50, 65, and 80 dB SPL. The ceiling is the loudness of unaided speech for a listener with typical hearing at that level, and is identical across profiles. The remainder of each bar's height to 100% is the budget available to purchase high-frequency audibility before any gain is prescribed. A 0 marks a level at which unamplified speech is inaudible to the model.

Figure 2. Change in optimized insertion gain when the minimum-gain floor is relaxed from 0 dB to -10 dB, at a 0.5-sone loudness budget, by audible edge frequency (anchor penalty on; all 16 pairs matched in loudness). Grey lines show individual audiograms; the black line is their mean. Because the paired solutions are at matched loudness, loudness given up at one frequency is spent at another.